

Amrit Aryal

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SUMMARY

Data Engineer with 4+ years of experience designing scalable ETL/ELT pipelines, data quality frameworks, and cloud-based analytics solutions across finance, healthcare, and higher education. Experienced with Spark, SQL, Python, Airflow, dbt, Snowflake, and GCP. Proven track record improving pipeline performance, automating workflows, and supporting enterprise-scale analytics.

SKILLS

Languages: SQL, Python, PySpark, Scala, Shell Script

Data Engineering Frameworks: Apache Spark, Apache Airflow, dbt, Kafka

Cloud & Databases: Snowflake, Databricks, PostgreSQL, MySQL, SQL Server, AWS, GCP, Azure

BI & Visualization: Tableau, Power BI

AI/ML: PyTorch, scikit-learn, LangChain, LangGraph

DevOps & Tooling: Docker, Git, CI/CD

Domains: ETL/ELT, Data Warehouse, Data Lake, Data Modeling, Batch Processing, Pipeline Orchestration, Data Quality, Data Lineage, Data Governance, Monitoring, Observability

EXPERIENCE

Data Engineer Associate

Jun 2025 – Aug 2026

Miami University IT Services, BI Unit

Oxford, OH

- Automated a critical daily reporting process by migrating Python scripts to a serverless GCP pipeline (Cloud Run, Cloud Scheduler), eliminating manual intervention and resolving failures caused by system availability.
- Built a Tableau executive dashboard for Solution Delivery throughput, using Python to validate a 680K-ticket dataset, map 10+ team names, and uncover data quality issues before visualization.
- Analyzed Oracle Analytic Server usage data to extract migration insights, supporting the university's transition from Banner to Workday on a modern analytics platform.

Graduate Teaching Assistant

Aug 2024 – May 2026

Miami University, Dept. of Computer Science and Software Engineering

Oxford, OH

- Graded assignments and guided students on software engineering course projects, providing feedback on design, code, and testing.
- Held office hours and mentored students on software engineering concepts, projects, and best practices.

Data Engineer

Apr 2024 – Jul 2024

Infinite Computer Solutions (formerly Cotiviti)

Kathmandu, Nepal

- Automated 5 healthcare data pipelines using Shell Scripts and PL/SQL, reducing manual processing time by 25% in a regulated compliance environment.
- Validated and integrated public and private data sources using SQL-based quality checks in PostgreSQL, ensuring 99.8% accuracy for US healthcare clients.
- Performed user acceptance testing on data integration workflows, verifying data accuracy and completeness against client requirements before production release.

Data Engineer

Jan 2022 – Apr 2024

ExtensoData (FISoft Group)

Kathmandu, Nepal

- Built scalable batch-processing ETL pipelines with Apache Spark and Apache Airflow, orchestrating data operations across 30+ financial institutions and improving processing speed by 40%.
- Owned data quality and monitoring for pipelines processing 10M+ daily records, maintaining a perfect zero-critical-incident production record across the entire tenure.
- Led a team of 10 engineers migrating from a traditional SQL/Pentaho stack to a Spark and Airflow big-data platform, defining pipeline architecture and orchestration standards.

- Designed and executed pipeline tests to validate accuracy, completeness, and performance across 30+ institutions, ensuring reliability before production deployment.
- Collaborated with data scientists to clean, structure, and transform transactional data, increasing ML model prediction accuracy by more than 20%.
- Documented pipeline architectures, data lineage, and operational runbooks to ensure reproducible workflows and consistent reporting standards.

Software Engineer Intern and Trainee

Jun 2021 – Jan 2022

Grafti Offshore

Kathmandu, Nepal

- Supported senior developers with design, coding, testing, and bug fixing for an ed-tech platform, including writing and executing test cases.

PUBLICATIONS

From Narratives to Reasoning: Can LLMs Generate Reasoning-Ready Logic Forms?

Amrit Aryal, Daniela Incelezan. Presented at the International Joint Conference on Learning & Reasoning (IJCLR 2026), Valencia, Spain, Sep 2026.

Assessing the Capability of Large Language Models in Generating ASP-Compatible Logic Forms from Human Narratives

Amrit Aryal. Master's Thesis, Miami University, published on OhioLINK (Advisor: Dr. Daniela Incelezan).

PROJECTS

E-commerce Platform with Recommendation Engine

2020

Laravel, Vue.js, MySQL, Stripe, Python, scikit-learn

github.com/amritmaniaryal/e-commerce_site

- Built a full-stack e-commerce platform with Stripe payments and a Python-based collaborative-filtering recommendation engine.

Road Network Skeletonization

2025

Python, PyTorch, osmnx, rasterio, scikit-image

- Built a PyTorch U-Net recovering road-centerline skeletons from noisy masks (OSM-derived synthetic training data). Outperforms classical skeletonization as mask noise increases (F1 0.85 vs 0.75) and transfers zero-shot to real road networks (Massachusetts Roads, F1 0.99).

Human Activity Recognition from Wearable Sensors

2025

Python, scikit-learn, PyTorch

- Developed ML models classifying human activities from body-worn sensor data, achieving approximately 90% accuracy.

Wildfire Detection and Vegetation Recovery Prediction

2025

Python, PyTorch, Remote Sensing

- Built ML models on satellite imagery to detect wildfire-affected areas and predict post-fire vegetation recovery.

EDUCATION

Master of Science in Computer Science

July 2026

Miami University

Oxford, OH

GPA: 4.00/4.00 | Research: Machine Learning, LLMs, Knowledge Representation

Bachelor of Science in Computer Science

Apr 2021

Tribhuvan University